

Mohammad Siddiquir Rahman

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Orlando, FL - 32826, USA

WORK AND RESEARCH EXPERIENCE

- The University of Central Florida** August 2023 - Current
Graduate Research Assistant Orlando, FL
 - Utilized the ArcGIS API for Python (arcpy raster) to automate the processing of high-resolution Land Use/Land Cover (LULC) datasets for statewide environmental modeling
 - Engineered a weather-based predictive framework for Florida's power grid, integrating multi-dimensional atmospheric data with infrastructure layers using Python (ArcPy, GeoPandas, Rasterio, Xarray)
 - Experienced in developing and deploying ArcGIS Dashboards to visualize real-time and historical data, providing stakeholders with intuitive map-based interfaces for informed decision making
- The University of Alabama** August 2021 - August 2023
Graduate Teaching Assistant Tuscaloosa, AL
 - Led lab sections for "Intro to GIS," teaching 60+ students fundamental concepts in ArcMap/ArcGIS Pro, coordinate systems, georeferencing, and geospatial visualization
 - Developed custom Python scripts (ArcPy) to automate repetitive geoprocessing tasks, significantly reducing data turnaround time for manuscript preparation
 - Produced high-quality, publication-ready thematic maps for peer-reviewed journals, ensuring strict adherence to spatial accuracy and design standards
 - Identify/troubleshoot and resolve routine GIS software, hardware, and network problems
- GIS E-Learning Center, BRUR** January 2020 - May 2021
Research Assistant Rangpur, Bangladesh
 - Converted historical hard-copy maps into digital vector formats using advanced georeferencing techniques in QGIS and ArcGIS
 - Produced high-quality, publication-ready thematic maps for peer-reviewed journals, ensuring strict adherence to spatial accuracy and design standards
 - Developed knowledge of data analysis and research techniques, methods, and processes

EDUCATION

- The University of Central Florida** August 2023 - Spring 2027
PhD in Water Resources Engineering Orlando, FL
 - GPA: 3.84/4.00
 - Relevant Courses: Environmental Informatics and Remote Sensing, Coastal System Analysis, Machine Learning, Data Mining, Water Resources in a Changing Environment, Statistical Analysis
- The University of Alabama** August 2021 - August 2023
Master's in Geography (GIS Focused) Tuscaloosa, AL
 - Grade: 3.88/4
 - Relevant Courses: Web GIS Remote Sensing Analytics, Environmental Data Analysis, GIS for Civil Engineers, Remote Sensing I-II, Research Methodology in GIS, Seminar in GIS, Quantitative Methods
- Begum Rokeya University** December 2019
Bachelor in Disaster Management (GIS Focused) Rangpur, Bangladesh
 - GPA: 3.38/4
 - Relevant Courses: Introduction to Climatology, Geo-Statistics, Practical (GIS and Remote Sensing), Basic Cartography and application of map, Practical (computer aided analysis and use of software)

SKILLS

- **Programming Languages:** Python, R, JavaScript, MATLAB
- **Database Systems:** MySQL, PostgreSQL, Microsoft SQL Server
- **Data Science & Machine Learning:** Apache Spark, Dask, OSMnx, Geopandas, Xarray, Scikit-learn, TensorFlow, PyTorch, Keras, PyCUDA
- **DevOps & Version Control:** Git, GitHub, GitLab, SourceTree
- **Specialized Area:** Hydrologic Modeling (HEC-RAS, HEC-HMS), Geospatial Data Analysis (ArcGIS Pro, ArcPy, ArcGIS Online, Python), Big Data Analytics, High Performance Computing, Parallel Computing, Machine Learning
- **Mathematical & Statistical Tools:** Spatial Statistics, Probability & Frequency Analysis, Time Series Analysis, Hypothesis Testing
- **Research Skills:** Literature Review, Research Design & Methodology, Data Collection & Management, Data Analysis & Interpretation, Communication & Dissemination

FUNDED RESEARCH PROJECTS

- **Clean, Affordable, and Resilient Energy Systems (CARES)** August 2023 – July 2025
Tools: HPC, Python, R, ArcGIS, PyTorch, TensorFlow 
 - Geospatial Predictive Modeling: Developed a high-resolution weather-to-outage modeling framework by integrating multi-scale atmospheric raster data with utility infrastructure layers.
 - Vulnerability Mapping: Performed spatial risk assessments to identify disadvantaged communities and vulnerable nodes within the power grid, using demographic and geographic indicators to drive resilience planning.
 - Spatio-Temporal Analysis: Conducted advanced analysis on the intersection of compound extreme weather events and grid failures in Florida, identifying geographic hotspots for proactive infrastructure hardening.
 - Big Data Processing: Managed large-scale geospatial datasets on High-Performance Computing (HPC) systems, optimizing the ingestion and preprocessing of complex environmental data.
- **The Megalopolitan Coastal Transformation Hub (MACH)** July 2025 – Present
Tools: HPC, Python, R, ArcGIS, PyTorch, TensorFlow, PyCUDA
 - Regional-Scale Power Modeling: Contributing to the Power System Modeling team by architecting wide-area spatial models to hindcast and project power outage risks across the US East Coast.
 - Probabilistic Spatial Frameworks: Developing sensitivity models to analyze how shifting extreme weather patterns impact grid reliability, utilizing a probabilistic approach to quantify geographic uncertainty.
 - GPU-Accelerated Geoprocessing: Implementing PyCUDA to accelerate the processing of massive coastal datasets, enabling rapid simulation of land-use impacts on infrastructure vulnerability.
 - Multi-Source Integration: Synthesizing disparate geospatial data, including sea-level rise projections, land cover, and transmission line topology to build a unified vision of future grid resilience.

CLASS PROJECTS

- **National Land Cover Dataset - Land Cover Ratio Analysis** 
Tools: Google Earth Engine, JavaScript
 - Developed a cloud-based framework to automate the extraction and processing of NLCD (National Land Cover Database) data at the county scale across the US.
 - Engineered an interactive GEE Dashboard that calculates land-cover ratios and generates on-the-fly CSV reports, reducing manual data processing time by 90%.
 - Implemented spatial aggregation scripts to handle multi-temporal raster datasets, streamlining environmental impact assessments for regional planning.
- **ArcGIS Dashboard of US Power Plants** [\[Dashboard Link\]](#)
Tools: ArcGIS Online, ArcGIS Dashboards
 - Engineered a dynamic ArcGIS Dashboard to visualize the distribution and operational capacity of US power plants across the United States.

- Synthesized and visualized energy data to create linked widgets, including categorized bar charts and percentage-of-total capacity pie charts.
- **Cartographic Design:** Applied custom symbology and cluster mapping to large-scale point datasets to ensure legibility and performant map rendering at multiple zoom levels.

ONLINE COURSES

ESRI Online Courses: Taken several online courses related to ArcGIS pro, ArcGIS online, ArcGIS WebApp, ArcGIS Arcade, Geodatabase etc.

LinkedIn Learning: Geospatial analysis in python, FME etc.

PUBLICATIONS

GOOGLE SCHOLAR | C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [C.1] **Rahman, M. S., Wahl, T., Nagaraj, M., & Rodriguez Enriquez, A. (2024). Compound Weather Extremes and Power Outages in Florida.** In *AGU Fall Meeting*, 2024, Washington, D.C.
- [J.1] **Rahman, M. S., Wahl, T., Nagaraj, M., & Enríquez, A. R. (Year). Characteristics of Power Outages from Compound Weather Extremes in Florida.** *Environmental Research: Climate* [In Press]
DOI:10.1088/2752-5295/adf252
- [J.2] Stevens, K. A., Aksha, S. K., **Rahman, M. S.**, Trader, E., Longenecker, H., Entress, R., Wahl, T., Emrich, C. T., & Davis, K. O. (2025). **Shedding light on inequities of power outages through data transparency.** *Progress in Energy*, 7(3), 033002. DOI:10.1088/2516-1083/addc30
- [J.3] Belligoni, S., Trader, E., Li, M., **Rahman, M. S.**, Ali, J., Enriquez, A. R., Nagaraj, M., Aksha, S. K., Stevens, K. A., Wahl, T., Emrich, C. T., Alam, N., Ganguly, A. R., Bukvic, A., Dinh, C., Pham, K. D., & Nguyen, T. D. (2025). **Transdisciplinary research promoting clean and resilient energy systems for socially vulnerable communities: A review.** *Renewable and Sustainable Energy Reviews*, 213, 115434. DOI:10.1016/j.rser.2025.115434
- [J.4] **Rahman, M. S., Senkbeil, J. C., & Keellings, D. J. (2023). Spatial and Temporal Variability of Extreme Precipitation Events in the Southeastern United States.** *Atmosphere*, 14(8), 1301. DOI:10.3390/atmos14081301
- [J.5] Musabbir, M., Islam, A. R. M. T., **Rahman, M. S.**, Pal, S. C., Alam, E., & Mallick, J. (2023). **Spatiotemporal variability of rainfall erosivity and its teleconnection with atmospheric circulation in monsoon-driven climate region.** *CATENA*, 222, 106762. DOI:10.1016/j.catena.2022.106762
- [J.6] Azad, M. A. K., Islam, A. R. M. T., Ayen, K., **Rahman, M. S.**, Shahid, S., & Mallick, J. (2022). **Changes in monsoon precipitation patterns over Bangladesh and its teleconnections with global climate.** *Theoretical and Applied Climatology*, 149(1–2), 651–669. DOI:10.1007/s00704-022-04021-w
- [J.7] Azad, M. A. K., Islam, A. R. M. T., **Rahman, M. S.**, & Ayen, K. (2021). **Development of novel hybrid machine learning models for monthly thunderstorm frequency prediction over Bangladesh.** *Natural Hazards*, 106(3), 2633–2659. DOI:10.1007/s11069-021-04532-5
- [J.8] **Rahman, M. S.**, Azad, M. A. K., Hasanuzzaman, M., Salam, R., Islam, A. R. M. T., Abdullah, K. P., Baten, S. M. A., Akter, T. T., & Hasan, S. M. D. (2021). **How air quality and COVID-19 transmission change under different lockdown scenarios? A case from Dhaka city, Bangladesh.** *Science of The Total Environment*, 754, 143161. DOI:10.1016/j.scitotenv.2020.143161
- [J.9] Islam, A. R. M., Ahmed, I., & **Rahman, M. S.** (2020). **Trends in cooling and heating degree-days overtimes in Bangladesh? An investigation of the possible causes of changes.** *Natural Hazards*, 104(2), 1397–1417. DOI:10.1007/s11069-020-04213-7
- [J.10] Islam, A. R. M. T., **Rahman, M. S.**, Khatun, R., & Hu, Z. (2020). **Spatiotemporal trends in the frequency of daily rainfall in Bangladesh during 1975–2017.** *Theoretical and Applied Climatology*, 141(3–4), 1523–1539. DOI:10.1007/s00704-020-03244-x
- [J.11] **Rahman, M. S.**, & Islam, A. R. M. T. (2019). **Are precipitation concentration and intensity changing in Bangladesh overtimes? Analysis of the possible causes of changes in precipitation systems.** *Science of the Total Environment*, 683, 127008. DOI:10.1016/j.scitotenv.2019.06.529

CONFERENCES

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| • AGU Annual Meeting 2024, Washington, D.C. | December 9-13, 2024 |
| • ASEMFL 2024 Annual Meeting, Orlando, FL | November 1-2, 2024 |
| • 49th Annual Natural Hazards Research Workshop, Boulder, CO | July 14-17, 2024 |
| • 2023 AAG Annual Meeting, Denver, CO | March 23-27, 2023 |

SUMMER SCHOOL EXPERIENCE

- **NSF I-GUIDE Summer School: Leveraging AI (funded)** August 5-9, 2024
Tools: Python, PyTorch, TensorFlow, GitHub
 - Developed a machine learning model using PyTorch to predict the efficacy of biochar in removing pharmaceutical waste from water sources.
 - Collaborated in a team to analyze results and presented key findings to researchers from NCAR and other universities.
- **MACH Summer School 2025** August 5-8, 2025
Networking, knowledge sharing, communication
 - Selected to participate in an intensive program featuring expert lectures, collaborative research discussions, and a technical field trip with fellow MACH members.

HONORS AND AWARDS

- **Graduate Dean’s Fellowship** August 2023
University of Central Florida
 - Awarded to superior graduate students newly enrolling in eligible programs based on academic excellence.
- **Faculty Cluster Initiative Fellowship** August 2025
University of Central Florida
 - Selected for fellowship supporting multidisciplinary research focused on solving complex, real-world problems.

PROFESSIONAL MEMBERSHIPS

- **American Geophysical Union** August 2024 - Present

ADDITIONAL INFORMATION

Research Interests: Hydro-climatic Extremes, Compound Extremes, Power Outage Modeling, Machine learning
General Interests: Data Analysis and Visualization, Badminton, Following tech updates

REFERENCES

1. **Dr. Thomas Wahl**
Associate Professor, Water Resource Engineering and UCF Coastal
University of Central Florida
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Relationship: Ph.D. Advisor
2. **Dr. Jason Senkbeil**
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The University of Alabama
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Relationship: Master’s Thesis Supervisor